

Nicholas Yeung

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EDUCATION

University of Wisconsin, Madison

Bachelor of Science in Computer Science & Data Science; GPA: 3.62

Madison, WI

Aug 2022 – May 2026

EXPERIENCE

Software Engineer Intern

June 2024 - August 2025

Cornell Communications

Milwaukee, WI

- Shipped a real-time floor plan and patient alert visualization feature in React and React Native using Mapbox GL, reducing average alert-response time by 25% (measured via call device logs) in nursing homes.
- Built an internal scheduling platform in Django and PostgreSQL integrated with Microsoft Outlook via the Graph API, supporting department-scoped event visibility, manager-approval workflows, and meeting room reservations.
- Built minimal Yocto Linux images bundling Asterisk and Node.js for on-site patient alert servers, cutting boot times and resource footprint versus legacy Ubuntu stack to enable migration to ARM SoC hardware.

PROJECTS

MottoNote | [Next.js/React](#), [TypeScript](#), [Firebase](#), [Cloud run](#), [Vertex AI](#), [LangChain](#)

- Built a production knowledge platform centralizing documents, emails, and notes across multiple SaaS sources, deployed to an enterprise pilot customer and validated against a synthetic dataset of 500+ PDFs and emails.
- Engineered a hybrid RAG pipeline combining dense vector search with sparse keyword matching, alongside a file-system-style retrieval API that exposes the knowledge base directly for LLM navigation.
- Architected a TypeScript backend with Cloud Run workers for document ingestion and embedding generation, shipping features via an AI-assisted workflow using Claude Code, Codex, and CodeRabbit.

Reels TUI | [Go](#), [FFmpeg](#), [ChromeDP](#) | [GitHub](#)

- Published an open-source TUI for browsing Instagram Reels with 5000+ installs across npm, Homebrew, and AUR, supporting likes, comments, bookmarking, sharing to DMs, and fully configurable keybinds.
- Reverse-engineered private GraphQL endpoints by intercepting network-layer requests via a ChromeDP backend.
- Built a media pipeline with audio clock A/V playback using statically linked FFmpeg decoders, backed by a FIFO disk cache with concurrent download deduplication and optimistic prefetching for seamless scrolling.

Piazza Bot | [Go](#), [Python](#), [Cassandra](#), [Kafka](#), [Redis](#), [Docker](#), [gRPC](#) | [GitHub](#)

- Built a distributed ingestion pipeline on a 2-node GPU cluster using Kafka for event-driven dispatch and Cassandra for dense and sparse vector storage, processing 700+ forum posts and 35+ hours of lecture recordings end-to-end.
- Designed a self-hosted QA pipeline where subagents evaluate expanded context windows in parallel, generating citations accurate to 20s of lecture recordings while rejecting the 85% of posts that are unanswerable.
- Developed a DP-based semantic chunking algorithm with $O(1)$ prefix-sum scoring, co-optimizing semantic coherence, chunk size, and fragmentation; published as a standalone Go library.

HoppyShare | [AWS](#), [Go](#), [Redis](#), [Kotlin](#), [Objective-C](#), [C++](#), [Flask](#), [Docker](#), [MQTT](#), [BLE](#) | [GitHub](#)

- Engineered a cross-platform clipboard and file sharing system with end-to-end encryption and mutual TLS authentication, supporting transfers across Windows, macOS, Linux, and Android.
- Built a modular backend with AWS Lambda for device on-boarding, a Mosquitto MQTT broker on EC2 with Flask API enforcing ACL and CRL-based access control, and Redis for session management.
- Developed native desktop clients in Go with C++ and Objective-C bindings for offline Bluetooth Low Energy fallback, with OS keychain integration and system tray background services.

AWARDS

1st Place at MadHacks 2025 | *3Docs* - 1st out of 114+ submissions, 380+ participants

Winner at Y Combinator Agent Jam 2025 | *Perp* - Track winner of invite-only AI agents hackathon at YC HQ

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Go, Python, C, Java, C#, SQL, HTML/CSS, R

Frameworks: React, Next.js, Node.js, Angular, React Native, Django, Flask, .NET, Tailwind

Infrastructure: Docker, AWS, Firebase, GCP, Azure, Kafka, Redis, Cassandra, MQTT

Developer Tools: Git, CI/CD (Github Actions), Linux, Vercel, Claude Code, VS Code